higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

FITTING AND TURNING
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
1.1.1 A
1.1.2 B
1.1.3 B
1.1.4 A
1.1.5 A

(5 × 2) (10)

1.2
A Shaft
B Plastic bushes
C Blotting paper
D Nut
E Outer flange (5 × 1) (5)

1.3
- Clean taps after use.
- Store taps in a container after use.
- Clean the tapping fluid from the workbench or floor.
- Ensure that metal shavings are cleaned and thrown in the metal-scrap bin.
- Use the correct tools for the job. (Any relevant 5 × 1) (5)

1.4
- Reduces friction
- Cools workpiece
- Cools cutting tool
- Prevents rust
- Washes away iron chips from workpiece
- Improves surface finish
- Helps dissipate high heat caused by friction
- Extends life of cutting tool
- Prevents build-up of material on cutting surface (Any relevant 5 × 1) (5)

[25]

QUESTION 2

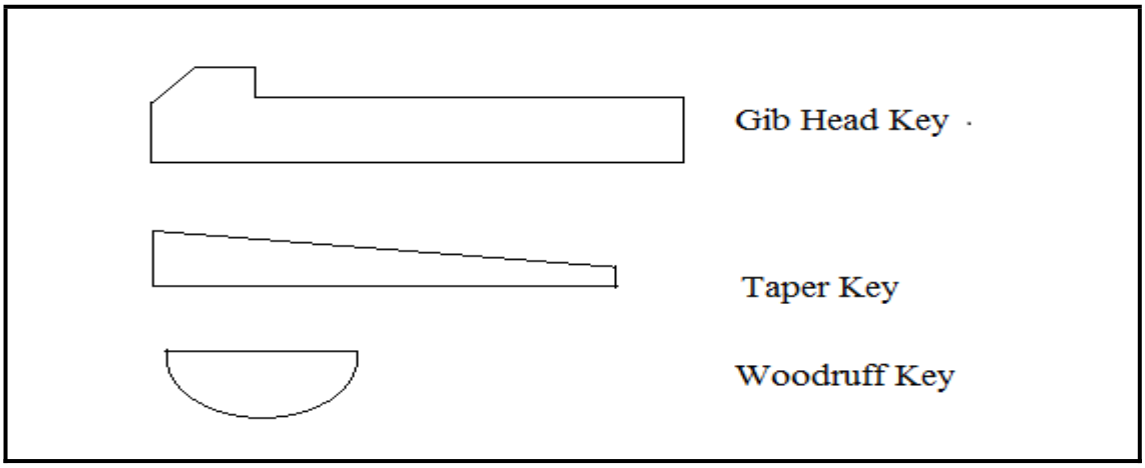
2.1
A Spindle motor
B Speed and feed gearbox
C Horizontal movement steering
D Base
E Pillar
F Elevating screw
G Elevating motor (7 × 1) (7)

2.2 D = 30 mm S = 50 m/min N = ?
$S = \pi \times D \times N$✓
$N = 50/\pi \times 30$✓✓
= 0,5307 r/min✓✓ (5)

2.3
- Check that the drive belts are in position and not damaged.
- Make sure that the power supply is secure and in the ON position.
- Ensure that the belt guard is closed.
- Check for the correct operation of the feed spindle.
- Make sure that the table can move freely and can lock.

(Any relevant 5 × 1) (5)

2.4



(3 × 2) (6)

2.5
- Thread-cutting screw
- Self-cutting screw
- Drive screw

(Any relevant 2 × 1) (2)
[25]

QUESTION 3

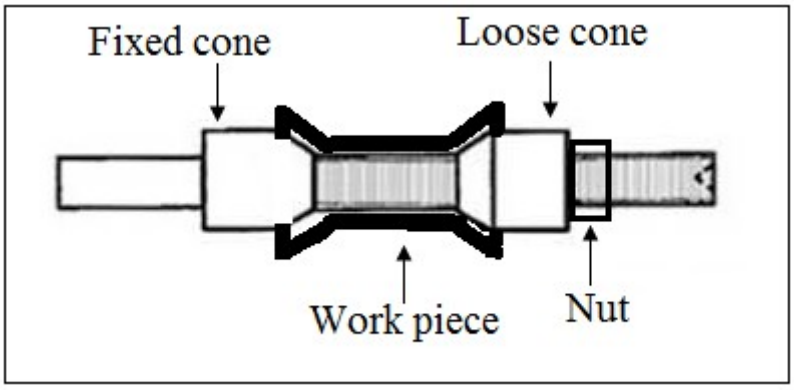
3.1
3.1.1 False
3.1.2 True
3.1.3 True
3.1.4 False
3.1.5 False

(5 × 1) (5)

3.2 Manual feed occurs when the operator of the lathe physically✓ makes adjustments while turning. Manual feed takes time✓ and the material removed leaves a rough and uneven surface/finish.✓ When the automatic feed is selected the machine makes adjustments automatically.✓ Automatic feed is quicker and the finish is more even and smooth.✓ (5)

3.3
- The setting time of the workpiece is time consuming.
- The chuck is heavy to mount on the lathe.
- The chuck is bulky.
- The strong gripping force of the jaws can damage workpieces.
- The chuck is difficult to set up, thus a dial-test indicator (DTI) must be used for accurate setting. (5)

3.4 Double-cone mandrel



(1 for mark + 4 for labelling) (5)

3.5
- The workpiece can get loose from the chuck.
- The cutting tool may break or become blunt.
- There might be play between the slides.
- There might be a lack of coolant.
- The centre hole may wear out.

(Any relevant 5 × 1) (5)
[25]

QUESTION 4

4.1
4.1.1 D
4.1.2 E
4.1.3 A
4.1.4 F
4.1.5 B
4.1.6 G
4.1.7 C
4.1.8 H

(8 × 1) (8)

4.2
- Eye protection
- Hairnet
- Gloves/Hand protection
- Suitable overalls
- SABS-approved safety boots

(Any relevant 5 × 1) (5)

4.3
- Speed
- Condition of machine
- Type of material being machined
- Condition of material
- Type of cutting tool

(Any relevant 5 × 1) (5)

4.4
- Ensure that there is an adequate supply of coolant.
- Make sure that the speed and feed is correct.
- Check that the workpiece is secure during cutting.
- Ensure that the cutting tips are sharp.
- When using automatic feed, ensure that the table trips are in position.

(Any relevant 5 × 1) (5)

4.5 The roughing cut is the first cut. Excessive material is removed and the workpiece is brought close to the required size.

(Any relevant answer) (2)

[25]

TOTAL: 100

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